



Course Instructor	Aneesa Nawaz, Asma Masood, Fahad Riaz, Jamil Usmani, Moheez-UR-Rahim, Shahid Ashraf, Urooj	Semester	Spring
Batch/Section(s)	Batch 2025 sections	Year	2026
Course Title	MT2008 Multivariate Calculus	Credit Hours	3
Prerequisite(s)	MT119- Calculus and Analytical Geometry	Course TA	

1 Course description and requisites

This course introduces calculus of functions of several variables with emphasis on geometric interpretation and applications. Topics include parametric and polar curves, partial derivatives, multiple integrals, vector calculus, and integral theorems. Students develop analytical skills necessary for solving problems arising in science, engineering, and computing.

2 Course objectives

Upon successful completion of this course, students will be able to:

- Analyze curves and surfaces using parametric and polar representations.
- Compute partial derivatives and apply gradient-based methods.
- Solve optimization problems using Lagrange multipliers.
- Evaluate double and triple integrals in various coordinate systems.
- Apply vector calculus theorems to physical and mathematical problems.

3 Grading Plan

1	Term-exams(M1)	15%
2	Term-exams(M2)	15%
3	Final Exam(F)	50%
4	Assignments	10 %
5	Quizzes	10%

Assigned Program Learning Outcome

No.	PLO Description	Level	Tools
02	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using the first principles of mathematics, natural sciences, and computing sciences.	R	

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, Pr=Presentation, P=Project, M = Midterm, F=Final.

Course Learning Outcome Statements

CLO	Course Learning Outcome Statements	Domain	Taxonomy level	PLO	Tools
01	Understand the basic concepts and know the basic techniques of differential and integral calculus of functions of several variables.	Cognitive	1	02	Q1, Q2, A1, A2, A3, M1, F
02	Apply the theory to calculate the gradients, directional derivatives, arc length of curves, area of surfaces, and volume of solids.	Cognitive	3	02	A1, M1, F
03	Solve problems involving maxima and minima, line integral and surface integral, and vector calculus.	Cognitive	4	02	Q3, A2, A3, M2, F
04	Apply gradient and derivative for solving various problems arising in sciences.	Cognitive	4	02	Q3, A2, A3, M2, F

Tools: $A=Assignment(A1,A2,A3)$, $M=Midterm(M1,M2)$, $F=Final$, $Q=Quiz(Q1,Q2,Q3)$

4 Standard Textbooks

- **James Stewart** , *Multivariable Calculus*, 9th Edition (Metric Version), Cengage Learning

Reference Books

1. Howard Anton et al., *Calculus*, John Wiley & Sons
2. Thomas & Finney, *Calculus*, Pearson
3. Gilbert Strang, *Calculus*, S. Chand

5 Expected student conduct

- Be in classroom on time.
- Turn off your cell phones or any other electronic devices before entering the class.
- Maintain the decorum of the class room all the time.
- Submit your assignments on time, no assignment will be accepted after the deadline.

Tentative Weekly Lectures Schedule:				
Week	Contents / Topics	Exercise	Questions	CLO
1	Parametric Equations and Curves: Curves Defined by Parametric Equations, Graphing Parametric Curves with Technology, Families of Parametric Curves Calculus with Parametric Curves : Tangents, Areas, Arc Length	10.1, 10.2	1-34 1-12,15-20,35-40	CLO2 A1
2	Polar Coordinates and Polar Curves : The Polar Coordinate System, Relationship between Polar and Cartesian Coordinates, Polar Curves, Symmetry Calculus in Polar Coordinates : Area, Arc Length, Tangents	10.3 10.4	1-50 1-44	
3	Derivatives and Integrals of Vector Functions: Derivatives, Differentiation Rules, definite integral. Arc Length and Curvature: Arc Length, The Arc Length Function, unit tangent and unit normal vectors, Curvature.	13.2 13.3	1-28,37-42 1-8,19-27	
4	Functions of Several Variables: Functions of Two Variables, Level Curves and Contour Maps, Functions of Three or More Variables Limits and Continuity: Limits of Functions of Two Variables, Properties of Limits, Continuity	14.1 14.2	1-36 1-30,41-50	CLO2 Q1
5	Partial Derivatives : Partial Derivatives of Functions of Two Variables, Higher Derivatives Tangent Planes: Equation of a Tangent Plane, Linear Approximations, Differentials	14.3 14.4	2,9-64 3-10,15-22,31-38	CLO1
6	Mid 1 Exam			
7	Chain Rule Case 1, case 2, General version, Implicit Differentiation Directional Derivatives and the Gradient Vector: Directional Derivatives, Gradient Vector, Maximizing the Directional Derivative, Tangent Planes to Level Surfaces,	14.5 14.6	1-37 1-52	CLO1 CLO4
8	Maxima, Minima Extreme value of the function of two variables. Absolute & Relative Extrema, Extreme Value theorem, The second order Partials test Lagrange Multipliers Gradient Descent Algorithm (Material will be provided)	14.7 14.8	5-22 3-16	CLO3 CLO4
9	Double Integrals Double Integrals over Rectangular and General Regions, General Regions: Changing the Order of Integration	15.1 15.2	15-37 1-28	CLO2
10	Double Integrals in Polar Coordinates: Double Integrals in Polar Coordinates, Applications of Double Integrals Density and Mass, Moments and Centers of Mass, Moment of Inertia	15.3 15.4	9-16 5-18	CLO3
11	Mid 2 Exam			
12	Triple Integrals Triple Integrals over Rectangular Boxes, Changing the Order of Integration Coordinate Systems Triple Integrals in Cylindrical Coordinates, Triple Integrals in Spherical Coordinates Change of Variables and Jacobians : Change of Variables in Double Integrals, Change of Variables in Triple Integrals	15.6, 15.7, 15.8 15.9	1-22 15.7: 31-32, 15.8: 43-45 11-16	CLO1, CLO2
14 ,15	Vector Fields (16.1) Line Integrals (16.2–16.3) Green's, Stokes', and Divergence Theorems (16.4–16.9)			CLO3
16	Final Exam			